

Be My Anesthesiologist: A Virtual Anesthesia Volunteer Network for Remote Care

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Background

Rural hospitals and military medical units frequently face a critical shortage of anesthesiologists. In the United States alone, projections indicate a deficit of over 7,000 anesthesiologists by 2033, with shortages particularly severe in rural areas [1]. Many rural hospitals rely entirely on Certified Registered Nurse Anesthetists (CRNAs) without any on-site anesthesiologists. In some states, CRNAs are the sole anesthesia providers in nearly all rural hospitals [1]. Similarly, frontline military field hospitals or combat surgical teams may lack a dedicated anesthesiologist, placing anesthesia care in the hands of CRNAs or general physicians working under extreme conditions [2]. These gaps lead to delays in urgent surgeries, limited pain management options, and elevated stress for on-site providers. Although telemedicine has begun to address some of these challenges through institutional tele-anesthesia programs [2], existing models are often rigid and unsuitable for smaller hospitals or ad-hoc emergencies at unconventional hours. There is a pressing need for a more agile, human-centered solution that can bring expert guidance to the bedside when it is needed most.

Methods

Inspired by the app Be My Eyes, which connects blind individuals with sighted volunteers through live video [3], I conceptualized a system called Be My Anesthesiologist. This platform would create a volunteer network of credentialed anesthesiologists who can offer real-time assistance remotely. When a provider in a rural or military setting faces a situation requiring anesthesiology input, they can send a help request via the app. That request is sent to multiple anesthesiologist volunteers, and any available physician can choose to respond. Unlike Be My Eyes, however, participation in this network requires a rigorous screening process. Only certified anesthesiologists who meet a standardized set of qualifications and pass a thorough credentialing process will be allowed to volunteer. This ensures that only experienced, verified professionals provide guidance, maintaining the highest standard of care. Through secure audio or video communication, the anesthesiologist can assess the patient's condition, view monitors, and provide live guidance on drug administration, airway management, crisis protocols, or equipment troubleshooting. This structure avoids placing responsibility on a single individual while still ensuring that expert help is just moments away. The entire process is voluntary, non-intrusive, and built to respect both the urgency of patient care and the schedules of busy clinicians.

Artificial intelligence can play a powerful supporting role in this system. Drawing inspiration from the partnership between Be My Eyes and OpenAI, which enabled blind and low-vision users to access GPT-4's multimodal capabilities through a feature called Be My AI, a similar approach could dramatically enhance remote anesthesiology support. In that model, GPT-4 interprets images, offers real-time guidance, and escalates to human volunteers when needed. Be My Anesthesiologist could adopt a parallel structure, using AI not as a replacement for human expertise, but as a dynamic first-line assistant and triage tool.

AI algorithms can rapidly evaluate incoming help requests, detect urgent trends in patient data, and assist with real-time monitoring, enabling both the local provider and the remote anesthesiologist to focus on informed decision-making [4]. For example, the AI might alert users to subtle deteriorations in BIS levels or oxygen saturation, suggest anesthetic drug dosing based on patient age, weight, and comorbidities, and auto-document relevant interventions to reduce provider burden [5]. During critical moments when no volunteer is immediately available, an AI agent, powered by systems like GPT-4, could offer protocol-based, situationally aware guidance to stabilize the patient until a certified anesthesiologist can join.

While AI will never replace the expertise or empathy of a trained physician, its ability to enhance safety, reduce cognitive overload, and provide intelligent decision support makes it a natural and powerful ally in high-stakes environments [5]. OpenAI’s success with Be My Eyes demonstrates how responsible, human-centered AI can scale care, and this same architecture could be adapted for anesthesia, helping save lives in operating rooms from rural hospitals to active combat zones.

Results

The prototype AI Anesthesia Dashboard provides a dynamic, real-time visualization of patient vitals in a simulated operating room environment. As shown in Figure 1, the interface continuously plots six key parameters: BIS level, heart rate, pulse pressure, SpO₂, end-tidal CO₂, and respiratory rate on separate axes for intuitive trend assessment. Abnormal BIS values trigger automated alerts that populate the “Live Alerts” panel. A patient status indicator ("STABLE" or "ALERT") provides a quick-glance summary, while interactive vaporizer sliders allow adjustment of isoflurane and sevoflurane concentrations.

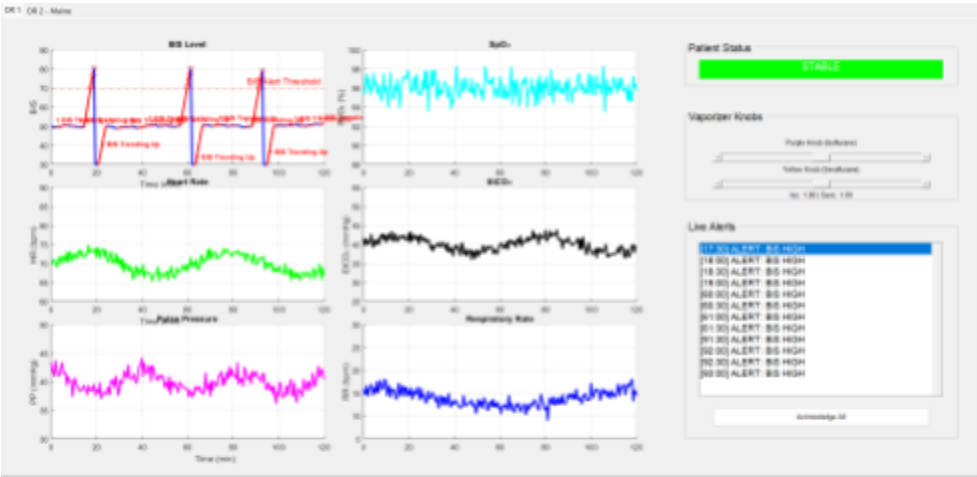


Fig. 1: Prototype AI Anesthesia Dashboard showing two hours of patient's vitals

The dashboard features a tabbed interface for monitoring multiple operating rooms simultaneously; tab labels (e.g., "OR 2 – Maine") reflect the geographic distribution of surgical sites, demonstrating how a single platform could enable anesthesiologists to supervise rural hospitals across the country. This responsive UI mirrors the complexity of real clinical environments and illustrates how an AI-enhanced monitoring and alerting system can support safe care delivery, especially when paired with remote anesthesiologist oversight. A demonstration of the prototype's functionality, including live vitals plotting, alert generation, and user interaction with the dashboard, is available in the following video: [Be My Anesthesiologist Demo Video](#)

Currently, vaporizer adjustments are achieved through hardware integration: the software communicates with a microcontroller (e.g., Arduino) that controls two servo motors. These motors are equipped with custom 3D-printed claws designed to grip and turn the physical vaporizer knobs on standard anesthesia machines. Future iterations will aim to eliminate the need for hardware by integrating directly with anesthesia systems through a proprietary digital interface, pending patent development.

Conclusion

This project aims to empower local clinicians, reduce unnecessary patient transfers, and ultimately save lives by extending anesthesiology expertise to places that need it most. The system allows anesthesiologists to participate on their terms while recognizing their contribution through salary bonus, CME credits, public acknowledgments, or charitable donations. Most importantly, it offers a way to create a lasting impact from anywhere in the world. This innovation has the potential to support rural health workers, military medics, and humanitarian responders who often work in isolation. By blending human generosity with emerging technologies it creates a model that could be adapted for other specialties in the future.

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